

1. Creating the Main Directory and Subdirectories:

`mkdir` = create directory

`-p` = create parent and child directories together

`{}` = create multiple directories at once

Subdirectories= documents,scripts,logs,backup

Poc:

```
(kali@kali)-[~]  
$ mkdir -p mid_exam/{documents,scripts,logs,backup}
```

Create employee_info.txt File:

```
(kali@kali)-[~]  
$ cd mid_exam/documents
```

Create the file and add company information:

```
(kali@kali)-[~/mid_exam/documents]  
$ echo "ABC Corporation - Employee Information" > employee_info.txt  
echo "Department: IT" >> employee_info.txt  
echo "Location: Dhaka, Bangladesh" >> employee_info.txt  
  
(kali@kali)-[~/mid_exam/documents]  
$ cat employee_info.txt  
ABC Corporation - Employee Information  
Department: IT  
Location: Dhaka, Bangladesh
```

Output:

```
ABC Corporation - Employee Information  
Department: IT  
Location: Dhaka, Bangladesh
```

Copy the File to Backup Directory and Rename It:

Poc:

```
(kali㉿kali)-[~/mid_exam/documents]
$ cp employee_info.txt ../backup/employee_backup.txt

(kali㉿kali)-[~/mid_exam/documents]
$
```

Explanation:

- cp = copy command
- .. = go one directory back

Verification:

```
(kali㉿kali)-[~/mid_exam/documents]
$ ls
employee_info.txt

(kali㉿kali)-[~/mid_exam/documents]
$
```

Create the script:

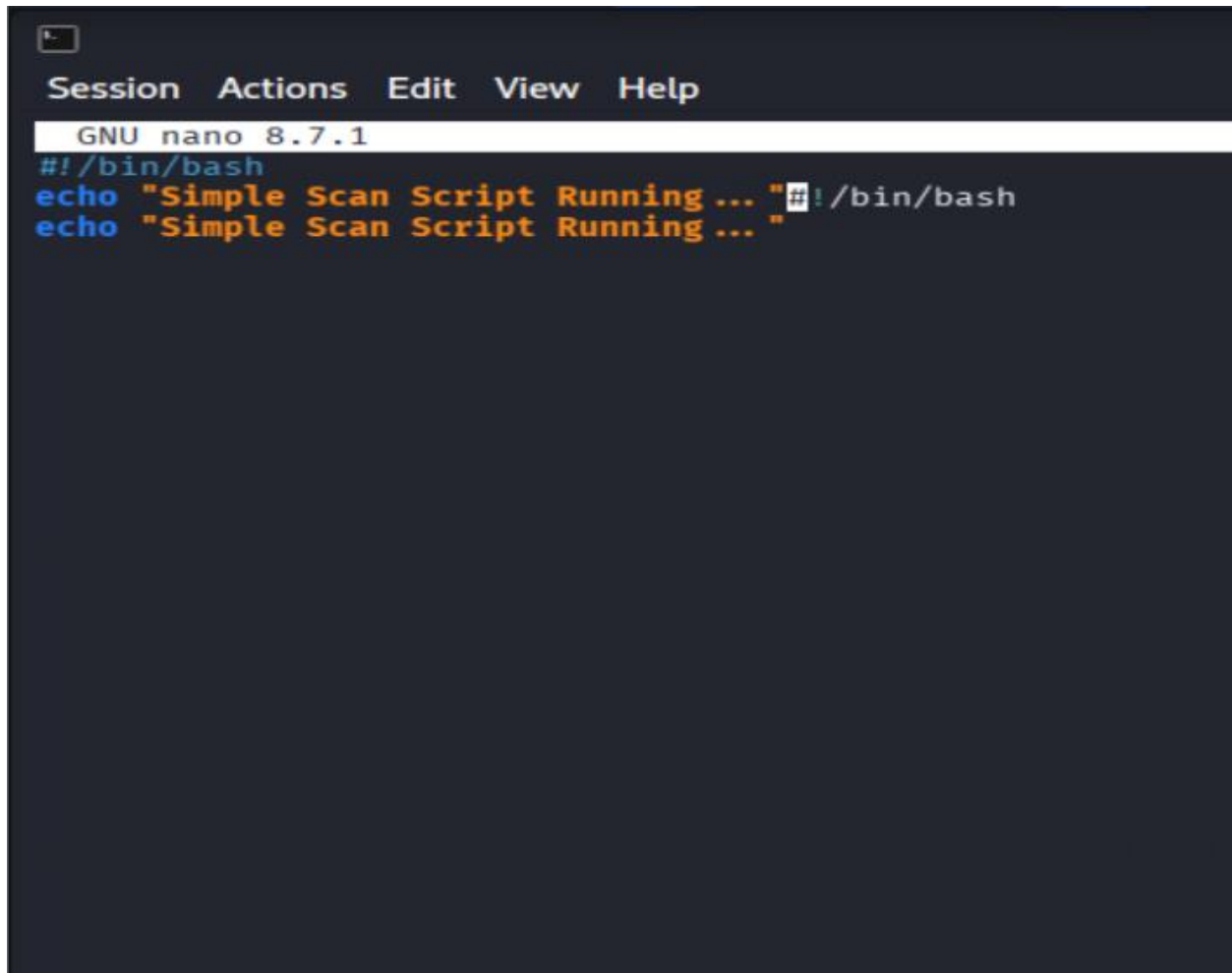
Command:

```
(kali㉿kali)-[~/mid_exam/documents]
$ nano scan.sh

(kali㉿kali)-[~/mid_exam/documents]
$
```

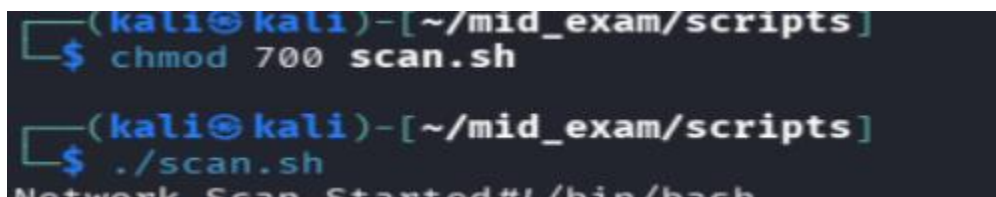
0 changes: 0 to add, 0 to delete, 0 to move, 0 seconds ago

Inside the file:



```
Session Actions Edit View Help
GNU nano 8.7.1
#!/bin/bash
echo "Simple Scan Script Running ... "
echo "Simple Scan Script Running ... "
```

Execute Permission Only to Owner:



```
(kali㉿kali)-[~/mid_exam/scripts]
$ chmod 700 scan.sh

(kali㉿kali)-[~/mid_exam/scripts]
$ ./scan.sh
Network Scan Started#!/bin/bash
```

Explanation:

7 = read + write + execute for owner

0 = no permission for group

0 = no permission for others

For Script Running:

```
(kali㉿kali)-[~/mid_exam/scripts]
$ ./scan.sh
```

Output:

```
$ ./scan.sh
Network Scan Started#!/bin/bash
Simple Scan Script Running ...
```

Collecting System Information:

Command for go to logs directory:

```
(kali㉿kali)-[~/mid_exam/scripts]
$ cd ../logs
```

For creating network_report.txt:

```
(kali㉿kali)-[~/mid_exam/logs]
$ echo "==== Network Report =====" > network_report.txt
```

For adding Ip address:

```
$ echo "==== Network Report =====" > network_report.txt

(kali㉿kali)-[~/mid_exam/logs]
$ echo "IP Address:" >> network_report.txt
hostname -I >> network_report.txt

(kali㉿kali)-[~/mid_exam/logs]
$
```

For adding Hostname:

```
(kali㉿kali)-[~/mid_exam/logs]
$ echo "" >> network_report.txt
echo "Hostname:" >> network_report.txt
hostname >> network_report.txt
```

For adding logged-in user:

```
(kali㉿kali)-[~/mid_exam/logs]
$ echo "" >> network_report.txt
echo "Logged-in User:" >> network_report.txt
whoami >> network_report.txt
```

For adding ping Test Result:

Command:

```
(kali㉿kali)-[~/mid_exam/logs]
$ echo "" >> network_report.txt
```

Output:

```
$ echo "" >> network_report.txt
echo "Ping Test Results:" >> network_report.txt
ping -c 4 8.8.8.8 >> network_report.txt
```

Description

- -c 4 = send 4 packets only

Basic Nmap scan:

Description

- Performs a basic TCP scan
- Saves output to scan_result.txt

Display Running Processes

Command for Show Running Processes:

```
(kali㉿kali)-[~]  
$ ps aux
```

Output:

```
(kali㉿kali)-[~]  
$ ps aux  
USER          PID %CPU %MEM    VSZ   RSS TTY      STAT START   TIME COMMAND  
root           1  0.0  0.3 24776 15368 ?        Ss   07:25   0:17 /sbin/init splash  
root           2  0.0  0.0      0     0 ?        S    07:25   0:00 [kthreadd]  
root           3  0.0  0.0      0     0 ?        S    07:25   0:00 [pool_workqueue_release]  
root           4  0.0  0.0      0     0 ?        I<   07:25   0:00 [kworker/R-rcu_gp]  
root           5  0.0  0.0      0     0 ?        I<   07:25   0:00 [kworker/R-sync_wq]  
root           6  0.0  0.0      0     0 ?        I<   07:25   0:00 [kworker/R-kvfree_rcu_reclaim]  
root           7  0.0  0.0      0     0 ?        I<   07:25   0:00 [kworker/R-slub_flushwq]  
root           8  0.0  0.0      0     0 ?        I<   07:25   0:00 [kworker/R-netns]  
root          10  0.0  0.0      0     0 ?        I<   07:25   0:00 [kworker/0:0H-events_highpri]  
root          12  0.0  0.0      0     0 ?        I    07:25   0:00 [kworker/u512:0-ipv6_addrconf]  
root          13  0.0  0.0      0     0 ?        I<   07:25   0:00 [kworker/R-mm_percpu_wq]  
root          14  0.0  0.0      0     0 ?        S    07:25   0:04 [ksoftirqd/0]  
root          15  0.3  0.0      0     0 ?        I    07:25   1:12 [rcu_preempt]  
root          16  0.0  0.0      0     0 ?        S    07:25   0:00 [rcu_exp_par_gp_kthread_worker/1]  
root          17  0.0  0.0      0     0 ?        S    07:25   0:00 [rcu_exp_gp_kthread_worker]  
root          18  0.0  0.0      0     0 ?        S    07:25   0:02 [migration/0]  
root          19  0.0  0.0      0     0 ?        S    07:25   0:00 [idle_inject/0]  
root          20  0.0  0.0      0     0 ?        S    07:25   0:00 [cpuhp/0]  
root          21  0.0  0.0      0     0 ?        S    07:25   0:00 [cpuhp/1]  
root          22  0.0  0.0      0     0 ?        S    07:25   0:00 [idle_inject/1]  
root          23  0.0  0.0      0     0 ?        S    07:25   0:02 [migration/1]  
root          24  0.0  0.0      0     0 ?        S    07:25   0:03 [ksoftirqd/1]  
root          26  0.0  0.0      0     0 ?        I<   07:25   0:00 [kworker/1:0H-events_highpri]  
root          27  0.0  0.0      0     0 ?        S    07:25   0:00 [cpuhp/2]  
root          28  0.0  0.0      0     0 ?        S    07:25   0:00 [idle_inject/2]  
root          29  0.0  0.0      0     0 ?        S    07:25   0:02 [migration/2]  
root          30  0.0  0.0      0     0 ?        S    07:25   0:02 [ksoftirqd/2]  
root          32  0.0  0.0      0     0 ?        I<   07:25   0:00 [kworker/2:0H-events_highpri]  
root          33  0.0  0.0      0     0 ?        S    07:25   0:00 [cpuhp/3]  
root          34  0.0  0.0      0     0 ?        S    07:25   0:00 [idle_inject/3]  
root          35  0.0  0.0      0     0 ?        S    07:25   0:02 [migration/3]  
root          36  0.0  0.0      0     0 ?        S    07:25   0:01 [ksoftirqd/3]  
root          42  0.1  0.0      0     0 ?        I    07:25   0:21 [kworker/u514:1-flush-8:0]  
root          45  0.0  0.0      0     0 ?        S    07:25   0:00 [kdevtmpfs]  
root          46  0.0  0.0      0     0 ?        I<   07:25   0:00 [kworker/R-inet_frag_wq]  
root          47  0.0  0.0      0     0 ?        I    07:25   0:00 [rcu_tasks_kthread]  
root          48  0.0  0.0      0     0 ?        S    07:25   0:00 [rcu_tasks_rude_kthread]
```

Description

- “ps aux” command Displays all running processes

Command for Monitor System Activity:

```
(kali㉿kali)-[~]  
$ top
```

Output:

```
top - 13:10:18 up 5:44, 1 user, load average: 0.44, 0.69, 0.65
Tasks: 269 total, 3 running, 266 sleeping, 0 stopped, 0 zombie
%Cpu(s): 10.4 us, 6.0 sy, 0.0 ni, 83.3 id, 0.0 wa, 0.0 hi, 0.4 si, 0.0 st
MiB Mem : 4171.5 total, 727.8 free, 1917.4 used, 1800.7 buff/cache
MiB Swap: 953.7 total, 953.7 free, 0.0 used, 2254.1 avail Mem
```

PID	USER	PR	NI	VIRT	RES	SHR	S	%CPU	%MEM	TIME+	COMMAND
969	root	20	0	517436	215716	103404	S	10.3	5.0	12:30.64	Xorg
2119	kali	20	0	1281316	134184	91588	S	5.8	3.1	10:21.64	xfwm4
2616	kali	20	0	918648	49084	36072	R	5.5	1.1	7:54.58	vmtoolsd
156011	kali	20	0	811660	70036	53004	S	3.9	1.6	0:07.00	qterminal
158656	kali	20	0	10460	5988	3808	R	3.5	0.1	0:02.72	top
475	root	20	0	253136	10092	8348	S	2.6	0.2	6:45.03	vmtoolsd
2197	kali	20	0	1201024	66556	33100	S	2.3	1.6	0:23.86	xfce4-panel
2281	kali	20	0	315484	67612	24912	S	2.3	1.6	5:42.07	wrapper-2.0
2283	kali	20	0	277136	30548	23228	S	1.9	0.7	4:23.33	wrapper-2.0
28939	kali	20	0	3205276	424956	183192	S	1.6	9.9	8:17.29	x-www-browser
964	root	20	0	1565576	46948	31516	S	1.3	1.1	3:25.70	containerd
29739	kali	20	0	2545692	177724	109792	S	1.3	4.2	1:51.19	Privileged Cont
154041	root	20	0	0	0	0	R	1.3	0.0	0:06.28	kworker/2:3+events
154058	root	20	0	0	0	0	I	1.0	0.0	0:05.45	kworker/0:0+events
28817	mysql	20	0	1523204	160740	30408	S	0.6	3.8	0:22.30	mariadb
45699	kali	20	0	811092	70496	52812	S	0.6	1.7	0:09.65	qterminal
141950	root	20	0	0	0	0	I	0.6	0.0	0:17.89	kworker/1:1+events_power_efficient
154022	root	20	0	0	0	0	I	0.6	0.0	0:03.63	kworker/3:3+events_power_efficient
15	root	20	0	0	0	0	I	0.3	0.0	1:13.95	rcu_preempt
1058	root	20	0	1678400	78708	52836	S	0.3	1.8	0:12.67	dockerd
3312	kali	20	0	811328	70176	53220	S	0.3	1.6	1:01.35	qterminal
125038	kali	20	0	811092	69056	52612	S	0.3	1.6	0:03.30	qterminal
158272	root	20	0	16528	6916	5852	S	0.3	0.2	0:00.04	systemd-userwor
159400	kali	20	0	3576	2204	2060	S	0.3	0.1	0:00.01	bwrap

Generate MD5 and SHA256 Hashes:

Command for move to logs directory if not already there:

```
(kali㉿kali)-[~]
$ cd ~/mid_exam/logs
```

Create hash report:

```
(kali㉿kali)-[~/mid_exam/logs]
$ echo "==== Hash Report =====> hash_report.txt

(kali㉿kali)-[~/mid_exam/logs]
$
```

Command for generate MD5 Hash:


```
(kali㉿kali)-[~/mid_exam/logs]
$ echo "" >> hash_report.txt
echo "MD5 Hash:" >> hash_report.txt
md5sum ../documents/employee_info.txt >> hash_report.txt

(kali㉿kali)-[~/mid_exam/logs]
$
```

Command for generate SHA256 Hash:

```
(kali㉿kali)-[~/mid_exam/logs]
$ echo "" >> hash_report.txt
echo "SHA256 Hash:" >> hash_report.txt
sha256sum ../documents/employee_info.txt >> hash_report.txt

(kali㉿kali)-[~/mid_exam/logs]
$
```

Investigate Authentication Logs:

Command for display last 15 line:

```
(kali㉿kali)-[~/mid_exam/logs]
$ sudo journalctl | tail -15
```

Output:

```
(kali㉿kali)-[~/mid_exam/logs]
$ sudo journalctl | tail -15
[sudo] password for kali:
May 12 13:14:46 kali systemd[1]: Starting NetworkManager-dispatcher.service - Network Manager Script Dispatcher Service ...
May 12 13:14:46 kali dbus-daemon[578]: [system] Successfully activated service 'org.freedesktop.nm_dispatcher'
May 12 13:14:46 kali systemd[1]: Started NetworkManager-dispatcher.service - Network Manager Script Dispatcher Service.
May 12 13:14:57 kali systemd[1]: NetworkManager-dispatcher.service: Deactivated successfully.
May 12 13:15:01 kali CRON[161685]: pam_unix(cron:session): session opened for user root(uid=0) by root(uid=0)
May 12 13:15:01 kali CRON[161687]: (root) CMD (command -v debian-sa1 > /dev/null && debian-sa1 1 1)
May 12 13:15:02 kali CRON[161685]: pam_unix(cron:session): session closed for user root
May 12 13:17:02 kali CRON[162666]: pam_unix(cron:session): session opened for user root(uid=0) by root(uid=0)
May 12 13:17:02 kali CRON[162668]: (root) CMD (cd / && run-parts --report /etc/cron.hourly)
May 12 13:17:02 kali CRON[162666]: pam_unix(cron:session): session closed for user root
May 12 13:25:02 kali CRON[166576]: pam_unix(cron:session): session opened for user root(uid=0) by root(uid=0)
May 12 13:25:02 kali CRON[166578]: (root) CMD (command -v debian-sa1 > /dev/null && debian-sa1 1 1)
May 12 13:25:02 kali CRON[166576]: pam_unix(cron:session): session closed for user root
May 12 13:26:28 kali sudo[167235]: kali : TTY=pts/4 ; PWD=/home/kali/mid_exam/logs ; USER=root ; COMMAND=/usr/bin/journalctl
May 12 13:26:28 kali sudo[167235]: pam_unix(sudo:session): session opened for user root(uid=0) by kali(uid=1000)

(kali㉿kali)-[~/mid_exam/logs]
$
```

Search for Keyword "failed":

```
(kali㉿kali)-[~/mid_exam/logs]
$ sudo journalctl | grep -i "failed"
May 04 06:22:05 kali kernel: pci 0000:0
```


Output:

```
sudo journalctl | grep -i failed
May 04 06:22:05 kali kernel: pci 0000:00:15.3: bridge window [io size 0x1000]: failed to assign
May 04 06:22:05 kali kernel: pci 0000:00:15.4: bridge window [io size 0x1000]: failed to assign
May 04 06:22:05 kali kernel: pci 0000:00:15.5: bridge window [io size 0x1000]: failed to assign
May 04 06:22:05 kali kernel: pci 0000:00:15.6: bridge window [io size 0x1000]: failed to assign
May 04 06:22:05 kali kernel: pci 0000:00:15.7: bridge window [io size 0x1000]: failed to assign
May 04 06:22:05 kali kernel: pci 0000:00:16.3: bridge window [io size 0x1000]: failed to assign
May 04 06:22:05 kali kernel: pci 0000:00:16.4: bridge window [io size 0x1000]: failed to assign
May 04 06:22:05 kali kernel: pci 0000:00:16.5: bridge window [io size 0x1000]: failed to assign
May 04 06:22:05 kali kernel: pci 0000:00:16.6: bridge window [io size 0x1000]: failed to assign
May 04 06:22:05 kali kernel: pci 0000:00:16.7: bridge window [io size 0x1000]: failed to assign
May 04 06:22:05 kali kernel: pci 0000:00:17.3: bridge window [io size 0x1000]: failed to assign
May 04 06:22:05 kali kernel: pci 0000:00:17.4: bridge window [io size 0x1000]: failed to assign
May 04 06:22:05 kali kernel: pci 0000:00:17.5: bridge window [io size 0x1000]: failed to assign
May 04 06:22:05 kali kernel: pci 0000:00:17.6: bridge window [io size 0x1000]: failed to assign
May 04 06:22:05 kali kernel: pci 0000:00:17.7: bridge window [io size 0x1000]: failed to assign
May 04 06:22:05 kali kernel: pci 0000:00:18.2: bridge window [io size 0x1000]: failed to assign
May 04 06:22:05 kali kernel: pci 0000:00:18.3: bridge window [io size 0x1000]: failed to assign
May 04 06:22:05 kali kernel: pci 0000:00:18.4: bridge window [io size 0x1000]: failed to assign
May 04 06:22:05 kali kernel: pci 0000:00:18.5: bridge window [io size 0x1000]: failed to assign
May 04 06:22:05 kali kernel: pci 0000:00:18.6: bridge window [io size 0x1000]: failed to assign
May 04 06:22:05 kali kernel: pci 0000:00:18.7: bridge window [io size 0x1000]: failed to assign
May 04 06:22:05 kali kernel: pci 0000:00:18.6: bridge window [io size 0x1000]: failed to assign
May 04 06:22:05 kali kernel: pci 0000:00:18.5: bridge window [io size 0x1000]: failed to assign
May 04 06:22:05 kali kernel: pci 0000:00:18.4: bridge window [io size 0x1000]: failed to assign
May 04 06:22:05 kali kernel: pci 0000:00:18.3: bridge window [io size 0x1000]: failed to assign
May 04 06:22:05 kali kernel: pci 0000:00:18.2: bridge window [io size 0x1000]: failed to assign
May 04 06:22:05 kali kernel: pci 0000:00:17.7: bridge window [io size 0x1000]: failed to assign
May 04 06:22:05 kali kernel: pci 0000:00:17.6: bridge window [io size 0x1000]: failed to assign
May 04 06:22:05 kali kernel: pci 0000:00:17.5: bridge window [io size 0x1000]: failed to assign
May 04 06:22:05 kali kernel: pci 0000:00:17.4: bridge window [io size 0x1000]: failed to assign
May 04 06:22:05 kali kernel: pci 0000:00:17.3: bridge window [io size 0x1000]: failed to assign
May 04 06:22:05 kali kernel: pci 0000:00:16.7: bridge window [io size 0x1000]: failed to assign
May 04 06:22:05 kali kernel: pci 0000:00:16.6: bridge window [io size 0x1000]: failed to assign
May 04 06:22:05 kali kernel: pci 0000:00:16.5: bridge window [io size 0x1000]: failed to assign
May 04 06:22:05 kali kernel: pci 0000:00:16.4: bridge window [io size 0x1000]: failed to assign
May 04 06:22:05 kali kernel: pci 0000:00:16.3: bridge window [io size 0x1000]: failed to assign
May 04 06:22:05 kali kernel: pci 0000:00:15.7: bridge window [io size 0x1000]: failed to assign
May 04 06:22:05 kali kernel: pci 0000:00:15.6: bridge window [io size 0x1000]: failed to assign
```

To direct input to this VM, move the mouse pointer inside or press Ctrl+G.

Create system_info.sh Script

Command for go to scripts directory:

```
(kali@kali)-[~/mid_exam/logs]
$ cd ~/mid_exam/scripts
(kali@kali)-[~/mid_exam/scripts]
$
```

Command for create script:

```
(kali㉿kali)-[~/mid_exam/logs]
$ cd ~/mid_exam/scripts

(kali㉿kali)-[~/mid_exam/scripts]
$ nano system_info.sh
```

Output:

```
GNU nano 8.7.1 system_
#!/bin/bash

echo "Current Date:"
date

echo ""

echo "Current User:"
whoami

echo ""

echo "Current Working Directory:"
pwd

echo ""

echo "Hostname:"
hostname
```

Make Script Executable:

```
(kali㉿kali)-[~/mid_exam/scripts]
$ chmod +x system_info.sh

(kali㉿kali)-[~/mid_exam/scripts]
$
```

Description

- +x = add execute permission

Command for Run the Script:

```
(kali@kali)-[~/mid_exam/scripts]  
$ ./system_info.sh
```

Output:

```
Current Date:  
Tue May 12 01:43:13 PM EDT 2026  
  
Current User:  
kali  
  
Current Working Directory:  
/home/kali/mid_exam/scripts  
  
Hostname:  
kali  
  
(kali@kali)-[~/mid_exam/scripts]  
$
```

Verification Commands

View Network Report:

```

(kali㉿kali)-[~/mid_exam/scripts]
$ cat ~/mid_exam/logs/network_report.txt
===== Network Report =====
IP Address:
192.168.10.128 172.17.0.1

Hostname:
kali

Logged-in User:
kali

Ping Test Results:
PING 8.8.8.8 (8.8.8.8) 56(84) bytes of data.
64 bytes from 8.8.8.8: icmp_seq=1 ttl=128 time=81.4 ms
64 bytes from 8.8.8.8: icmp_seq=2 ttl=128 time=114 ms
64 bytes from 8.8.8.8: icmp_seq=3 ttl=128 time=145 ms

— 8.8.8.8 ping statistics —
4 packets transmitted, 3 received, 25% packet loss, time 3005ms
rtt min/avg/max/mdev = 81.420/113.490/145.405/26.122 ms

(kali㉿kali)-[~/mid_exam/scripts]
$ █

```

View scan result:

```

(kali㉿kali)-[~/mid_exam/scripts]
$ cat ~/mid_exam/logs/scan_result.txt
Starting Nmap 7.99 ( https://nmap.org ) at 2026-05-12 12:52 -0400

(kali㉿kali)-[~/mid_exam/scripts]
$ █

```

View hash report:

```

(kali㉿kali)-[~/mid_exam/scripts]
$ cat ~/mid_exam/logs/hash_report.txt
===== Hash Report =====

MD5 Hash:
5fa74cd067128a150ed48943cd5fc27f  ../documents/employee_info.txt

SHA256 Hash:
88b4a6a0a3c5644bc567ce0e180d02dde236ecc5d6df4ef97b49caa82e97cb0  ../documents/employee_info.txt

(kali㉿kali)-[~/mid_exam/scripts]
$ █

```

o direct input to this VM, move the mouse pointer inside or press Ctrl+G.

Check File Permissions

```
(kali㉿kali)-[~/mid_exam/scripts]
$ ls -l ~/mid_exam/scripts
total 8
-rwx----- 1 kali kali  88 May 12 12:07 scan.sh
-rwxrwxr-x 1 kali kali 161 May 12 13:38 system_info.sh

(kali㉿kali)-[~/mid_exam/scripts]
$
```